

Acute myeloid leukemia

≥20% myeloid blasts — or any blast % with a defining genetic lesion · classification and treatment are now genetically driven · every new diagnosis is triaged for the three emergencies before a regimen is chosen

Compiled from the Encyclopedia AML review note and Fellowship Anki deck · ELN 2022 · WHO 2022 / ICC · NCCN.

DIAGNOSIS

- ≥20% myeloid blasts — or any blast % with a defining fusion: t(8;21), inv(16)/t(16;16), PML::RARA. WHO 2022 also drops it for NPM1 but **keeps ≥20% for CEBPA-bZIP; ICC sets ≥10% for both**
- Also AML at <20%: **myeloid sarcoma** / extramedullary disease
- “Blasts” on smear is morphologic — **confirm lineage by flow**
- Workup: marrow aspirate + biopsy with **flow, karyotype + FISH, NGS**; CBC/smear (Auer rods); **coags + fibrinogen + D-dimer**; TLS labs; **baseline echo**; HLA typing early; LP only for CNS symptoms (consider if monocytic), after coags corrected
- **Stat FISH/PCR for PML::RARA** on any suspicion of APL
- Median age ~68. The three triage questions on every new AML: **Is it APL? Is there leukostasis? Is there tumor lysis?**

ELN 2022 GENETIC RISK

Risk	Representative genetics
Favorable	t(8;21) RUNX1::RUNX1T1 · inv(16)/t(16;16) CBFB::MYH11 (core-binding factor) · mutated NPM1 without FLT3-ITD · in-frame bZIP CEBPA
Intermediate	NPM1 + FLT3-ITD · wild-type NPM1 + FLT3-ITD · t(9;11) MLLT3::KMT2A · anything not otherwise classified
Adverse	Complex or monosomal karyotype · -5/del(5q), -7, abn(17p) · TP53 · MDS-related genes (ASXL1, BCOR, EZH2, RUNX1, SF3B1, SRSF2, STAG2, U2AF1, ZRSR2) · KMT2A rearrangements except t(9;11) and KMT2A-PTD · t(6;9), inv(3)

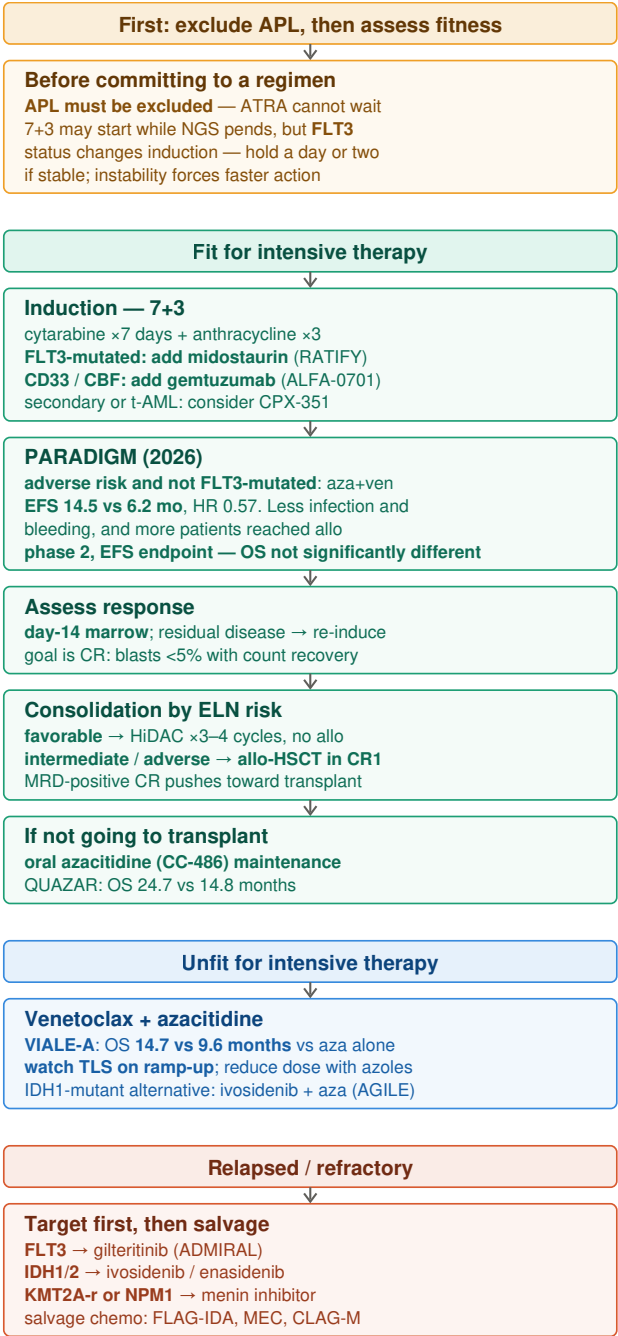
ELN 2022 changes worth knowing: FLT3-ITD is intermediate regardless of allelic ratio (ratio dropped) · only in-frame bZIP CEBPA is favorable · **NPM1 with adverse cytogenetics is reclassified adverse** · KIT co-mutation worsens CBF-AML prognostically, but ELN 2022 kept it favorable.

GENOTYPE → PHENOTYPE

FLT3-ITD	Hyperleukocytosis; normal karyotype. ↑ CNS risk , most when NPM1 co-mutated
KMT2A / 11q23	Monocytic — gum hypertrophy, leukemia cutis. Highest multivariable odds for CNS disease (OR ~7.6)
t(8;21)	Myeloid sarcoma ; aberrant CD19 / CD56 by flow
inv(16)	Abnormal marrow eosinophils with coarse basophilic granules (FAB M4Eo)
TP53	Complex karyotype, t-AML; venetoclax-resistant , poor outcome even after allo
DTA	DNMT3A, TET2, ASXL1 persist in CR as clonal hematopoiesis — not MRD markers

CNS disease is uncommon — ~0.6% at diagnosis, ~3% at relapse. Risk factors differ by series: SAL (n=3,261) found **FLT3-ITD ± NPM1**; another found **11q23 (OR 7.6)** and **WBC ≥100k** on multivariable analysis.

TREATMENT PATHWAY



REGIMENS & AGENTS

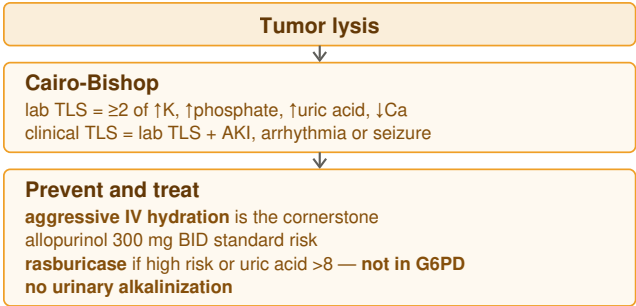
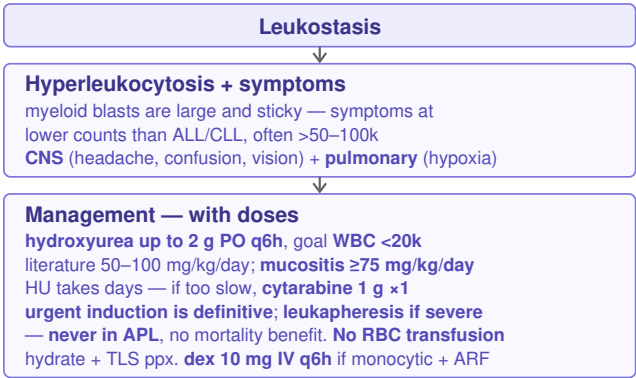
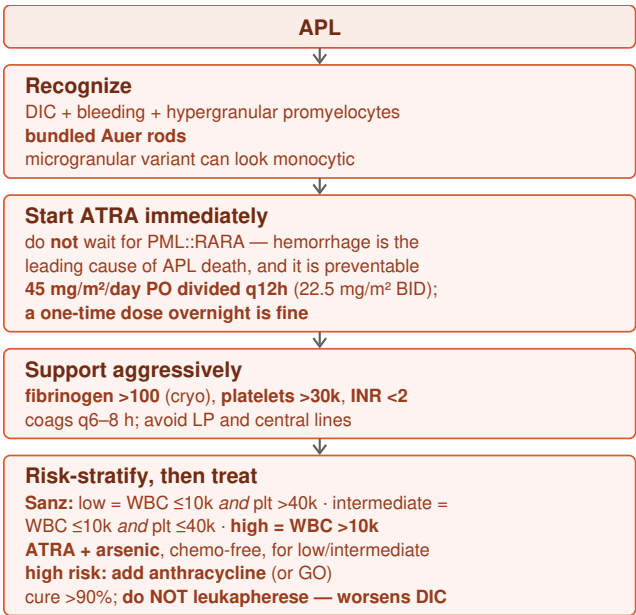
Regimen	Notes
7+3	Cytarabine infusion ×7 days + an anthracycline ×3 — the backbone for fit patients. Daunorubicin 60 mg/m² is the modern standard : E1900 showed 90 > 45, but AML17 found 90 vs 60 gave no CR or OS benefit and doubled 60-day mortality
+ midostaurin	FLT3-mutated (RATIFY). Quizartinib for FLT3-ITD (QuANTUM-First)
+ gemtuzumab	Fractionated dosing (ALFA-0701 — no CD33 cutoff). CBF benefit comes from the Hills meta-analysis , not ALFA-0701
CPX-351	Liposomal cytarabine + daunorubicin for t-AML / AML-MRC. Caveat : comparator was 7+3, not ven/aza — niche today is the fitter older adult bridging to allo
HiDAC	3 g/m² q12h on days 1, 3, 5 ×3–4 cycles; best in CBF. Age >60 → IDAC 1.5 g/m² — 3 g/m² is inappropriate (ocular, cerebellar, infectious toxicity). Renal dose-reduce
Venetoclax + azacitidine	Unfit patients (VIALE-A). Ramp-up with TLS prophylaxis; reduce venetoclax with CYP3A inhibitors

This table is **when and why**. Every regimen's dosing lives on page 3 — the two HiDAC numbers stay here because the dose *is* the teaching point.

MRD AND THE TRANSPLANT DECISION

- **Measure MRD at the end of every cycle** and again before transplant — order **HemeSTAMP MRD** (NGS) + **multiparameter flow cytometry**; add **NPM1 PCR** where applicable
- **MRD-positive CR is not CR for planning purposes** — it predicts relapse and moves an otherwise favorable-risk patient toward transplant
- **Allo-HSCT in CR1**: generally **not** for favorable risk · **yes** for intermediate and adverse risk · modified by MRD, age, comorbidity, and donor availability
- Allo cures through **graft-versus-leukemia**. **Auto-HSCT lacks GVL** and relapses more — niche for MRD-negative CR1 without a donor
- Fitness for intensive therapy is a composite of age, performance status and comorbidity (HCT-CI) — not age alone

THE THREE EMERGENCIES



DRUG TOXICITY & MONITORING

Agent	Watch for
Cytarabine	Pyrimidine analog. High-dose: cerebellar toxicity — cerebellar exam before every HiDAC dose, hold if abnormal; conjunctivitis → prophylactic steroid eye drops ; ara-C syndrome (fever, myalgia, rash)
Anthracyclines	Cumulative, dose-dependent cardiotoxicity — track total dose, baseline ± serial LVEF. Vesicant — central access
Venetoclax	Tumor lysis on ramp-up ; neutropenia; dose-reduce with azole antifungals (CYP3A)
Azacitidine vs decitabine	Both hypomethylating; azacitidine incorporates into RNA and DNA , decitabine DNA-only. Azacitidine: injection-site reactions
Gemtuzumab	Anti-CD33 calicheamicin ADC — hepatotoxicity and VOD/SOS , especially peri-transplant; fractionated dosing lowers risk
Arsenic trioxide	QT prolongation — monitor ECG, replete K and Mg
Differentiation syndrome	ATRA/ATO and also IDH and menin inhibitors. Fever, dyspnea, hypoxia, infiltrates, effusions, weight gain, AKI → dexamethasone 10 mg IV BID at first suspicion
Menin inhibitors	Revumenib, ziftomenib — differentiation syndrome + QT prolongation

TRIAL EVIDENCE

RATIFY	Midostaurin + 7+3 in FLT3-mutated — survival benefit
QuANTUM-First	Quizartinib + 7+3 in FLT3-ITD
ALFA-0701	Fractionated gemtuzumab + 7+3 — EFS/RFS benefit held in the final analysis but OS was no longer significant . Hills meta-analysis (n=3,325) is what shows the favorable-cytogenetics benefit: 5-yr OS +20.7%
E1900	Daunorubicin 90 vs 45 mg/m² — better CR and OS in age <60
AML17	Daunorubicin 90 vs 60 mg/m² — no CR or OS difference , and 60-day mortality 10% vs 5% (HR 1.98). Read with E1900: 60 is standard, not 90 (Burnett, Blood 2015)
VIALE-A	Azacitidine + venetoclax vs azacitidine — OS 14.7 vs 9.6 months (DiNardo, NEJM 2020)
PARADIGM	Aza+ven vs intensive induction in fit patients (CBF/FLT3/ NPM1 <60 excluded, 72% ELN adverse): EFS 14.5 vs 6.2 mo , HR 0.57; less infection and bleeding. Phase 2; OS not significant (Fathi, NEJM 2026)
AGILE	Ivosidenib + azacitidine in IDH1-mutant unfit patients
ADMIRAL	Gilteritinib in relapsed FLT3-mutated AML — survival benefit
SORMAIN	Sorafenib maintenance after allo in FLT3-ITD AML — 24-mo RFS 85% vs 53% , HR 0.39. Phase 2, n=83; off-label (Burchert, JCO 2020)
MORPHO	Gilteritinib maintenance after allo in FLT3-ITD, n=356. RFS missed overall (HR 0.68, P=0.052) — but benefit in the half who were MRD-positive pre- or post-HCT (HR 0.52, P=0.007) and none if MRD-negative . An MRD-guided decision (Levis, JCO 2024)
APL0406	ATRA + arsenic beat ATRA + chemo in low/intermediate-risk APL (Lo-Coco, NEJM 2013)
QUAZAR AML-001	Oral azacitidine (CC-486) maintenance in CR1 — OS 24.7 vs 14.8 months
AML15	IDAC vs HiDAC consolidation — no overall survival difference
SWOG S1203	IA regimen showed no benefit and more toxicity — 7+3 remains standard

GENETICS — FINE POINTS

inv(3) / t(3;3)	MECOM (EVI1, 3q26) activation; classically normal or high platelets ; adverse and venetoclax-resistant
t(6;9)	DEK::NUP214 — associates with FLT3-ITD and basophilia ; adverse
NPM1 flow mimic	NPM1-mutated AML can be HLA-DR– / CD34– and mimic APL — confirm PML::RARA
Therapy-related	After alkylators/RT : del(5q)/del(7q) ± TP53, latency ~5–7 yr. After topoisomerase-II: 11q23 KMT2A , latency ~2–3 yr
Germline	DDX41, RUNX1, CEBPA, GATA2 , telomere biology — send germline testing if young, family history, or multiple cytopenias; it changes related-donor selection

Acute myeloid leukemia

FOUR WAYS AN ACUTE LEUKEMIC DIES IN THE ED

- 1. Leukostasis — CVA, MI, respiratory failure
- 2. Bleeding / DIC / thrombosis
- 3. Infection — very low threshold for broad-spectrum antibiotics
- 4. Tumor lysis — renal failure, electrolyte derangement

OVERNIGHT DECISIONS

- 1. Triage floor vs ICU — leukapheresis and CRRT both require ICU
- 2. Rule out APL — start ATRA on any suspicion, do not wait for PML::RARA
- 3. Cyto reduction — see doses at right
- 4. TLS — start fluids + allopurinol, set lab frequency (q4h to daily). Rasburicase if uric acid >8: 0.15–0.2 mg/kg IV, in adults usually a single dose capped at 6 mg (label allows 0.2 mg/kg daily ×5 days). Check G6PD — contraindicated if deficient

SUPPORTIVE CARE

- Transfusion: platelets >10k (>20–50k if bleeding, procedure, or APL); irradiated and leukoreduced products
- Prophylaxis during induction: levofloxacin 500 mg PO daily · posaconazole DR 300 mg BID ×1 day, then 300 mg daily · acyclovir 400 mg PO BID (HSV-seropositive)
- Posaconazole is a strong CYP3A4 inhibitor — that is what forces the venetoclax reduction: 400 mg → 70 mg daily with posaconazole (100 mg with other strong inhibitors such as voriconazole), during ramp-up and beyond
- Hepatitis B: screen HBsAg + anti-HBc before therapy. HBsAg-positive → entecavir or tenofovir (preferred over lamivudine) for any systemic anticancer therapy, continued ≥12 months past the last cycle. Isolated anti-HBc-positive → prophylax or monitor HBV DNA closely; AML chemotherapy carries reactivation risk similar to lymphoma
- Febrile neutropenia: fever ≥38.3°C once (or ≥38.0°C for 1 h) with ANC <500 → cefepime 2 g IV q8h within 1 hour. Add vancomycin for hemodynamic instability, line infection, skin/soft tissue infection, pneumonia or known MRSA. Severe mucositis is a softer trigger than it looks — IDSA lists it only when fluoroquinolone prophylaxis was used and ceftazidime is the empiric agent (viridans strep); cefepime already covers viridans strep. Antifungal if fever persists >4–7 days
- Fertility counseling before chemotherapy; central venous access; early palliative care in older or adverse-risk disease

WHAT TO SEND

Test	When
Peripheral studies	Flow cytometry, AML prognosis assay (NPM1, FLT3-ITD, FLT3-TKD), IDH mutation analysis. Send if WBC >10k and marrow will be delayed, or the patient is old enough that you would rather not subject them to a marrow
Cytogenetics	FISH for PML::RARA and BCR::ABL1. Send from peripheral blood if there is any concern for APL or a delay in getting marrow
HLA typing	Send if the patient is a possible allogeneic transplant candidate

CYTOREDUCTION OVERNIGHT

Dose and trigger	
Hydroxyurea	Consider if WBC >20k. Overnight you can order a one-time dose. Typical range 500 mg–2 g, max 2 g q6h. Literature range 50–100 mg/kg/day; grade 3–4 mucositis rises at ≥75 mg/kg/day, so 2 g q6h is the ceiling, not the starting dose
Leukapheresis	Consider for evidence of leukostasis — headache, blurry vision, dyspnea/hypoxia, chest pain — typically with blasts >50k in AML (rare in ALL). Page apheresis 12027. Ask ER or the MICU triage fellow for a trialysis line; admit to MICU. Ask the apheresis RN to save the product in the FGR fridge and coordinate with the research team. Never in APL
Cytarabine	1 g ×1 may be considered for leukostasis when apheresis will be delayed

APL OVERNIGHT

- Start ATRA on any concern for APL — do not wait for PML::RARA
- Smear: bilobed / folded nuclei ± Auer rods. Page pathology on call (12005) if unsure
- ATRA 45 mg/m²/day PO divided q12h (= 22.5 mg/m² BID) — overnight you can give a one-time dose
- DIC labs q6–8h. Transfuse platelets >30, fibrinogen >100, INR <2

REGIMEN DOSING

Regimen	Dosing
7+3	Cytarabine 100–200 mg/m²/day IV continuous infusion ×7 days from day 1 (SHC uses 100) + daunorubicin 60 mg/m² IV daily days 1–3 schedule E1900; 60 vs 90 from AML17
7+3 + GO	Add gemtuzumab ozogamicin 3 mg/m² (max 4.5 mg) IV over 2 h on days 1, 4, 7 ALFA-0701 / Lambert 2019
7+3 + midostaurin	Add midostaurin 50 mg PO q12h on days 8–21 RATIFY / Stone 2017
CPX-351	Liposomal daunorubicin 44 mg/m² + cytarabine 100 mg/m² IV on days 1, 3, 5 Lancet 2018
FLAG-IDA-VEN	Fludarabine 30 mg/m² + cytarabine 1.5–2 g/m² IV days 2–6; idarubicin IV (ND-AML 8 mg/m² days 4–6; R/R 6 mg/m² days 4–5); filgrastim 5 mcg/kg days 1–7; venetoclax ramp-up (100/200/400 mg) days 1–14 DiNardo 2021
CLAD / LDAC / VEN	Cladribine 5 mg/m² IV days 1–5 + cytarabine 20 mg/m² (SQ BID days 1–10; 20 mg/m² IV daily preferred at SHC) + venetoclax PO days 1–21, alternating with azacitidine Kadia 2022
MEC	Mitoxantrone 6–8 mg/m² + etoposide 80–100 mg/m² + cytarabine 1 g/m², all daily days 1–5 (some series run 6 days) — dosing varies by protocol, check yours relapsed / refractory
CLAG-M	Cladribine 5 mg/m² IV days 1–5 + cytarabine 2 g/m² IV days 1–5 + mitoxantrone 10 mg/m² IV days 1–3 + G-CSF 300 mcg days 0–5 relapsed / refractory
HMA / VEN	Venetoclax (ramp-up) daily ×28 days + decitabine 20 mg/m² days 1–5 or azacitidine 75 mg/m² days 1–7 VIAL-E-A / DiNardo 2020

APL INDUCTION — DOSING

Regimen	Dosing
ATO / ATRA	Arsenic trioxide 0.15 mg/kg IV over 2 h daily from day 1 + ATRA 22.5 mg/m² PO BID from day 1 — both until hematologic CR or max 60 days APL0406 / Lo-Coco 2013
ATO / ATRA / IDA	ATO 0.15 mg/kg IV days 9–36 + ATRA 22.5 mg/m² BID days 1–36 + idarubicin on days 2, 4, 6, 8 by age: ≤60 yr 12 mg/m² · 61–70 yr 9 mg/m² · >70 yr 6 mg/m² APML4 / Iland 2012
ATO / ATRA / GO	ATO 0.3 mg/kg IV daily days 1–5, then 0.25 mg/kg twice weekly weeks 2–8 + ATRA 22.5 mg/m² BID from day 1 until CR or max 60 days + gemtuzumab 6 mg/m² IV once on day 1 — high risk only (WBC >10k) AML17 APL / Burnett 2015